

BARCELONA AP, Spain

WMO: 081810

Lat: 41.2928N

Lon: 2.0700E

Elev: 12

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.4	37.6	15.7	12.2	43.7	21.4	16.1	43.8	26.0	54.1	22.9	53.1	11.9	350	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.6	87.6	74.3	85.6	74.1	84.0	73.6	77.7	84.6	76.3	83.2	75.1	82.0	10.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	133.1	82.8	73.8	126.2	81.6	72.7	121.4	80.8	41.0	84.6	39.7	83.2	38.5	82.3	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	19.7	17.4	DB	31.9	91.0	3.0	2.8	29.7	93.0	28.0	94.6	26.3	96.2	24.2	98.2	(4)
(5)			WB	27.7	79.1	3.4	2.4	25.3	80.8	23.3	82.2	21.4	83.5	18.9	85.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.4	49.9	50.5	54.2	58.3	64.2	71.6	76.6	77.4	71.8	65.6	56.3	51.0	(6)
(7)		DBStd	10.72	4.48	4.56	3.82	3.58	4.04	4.50	3.61	3.60	4.08	4.54	5.19	4.86	(7)
(8)		HDD50	164	56	43	10	0	0	0	0	0	0	0	10	45	(8)
(9)		HDD65	2222	468	407	335	201	63	3	0	0	3	47	262	433	(9)
(10)		CDD50	4677	53	57	140	250	440	648	824	850	653	484	200	78	(10)
(11)		CDD65	1258	0	0	0	1	37	201	359	386	206	66	2	0	(11)
(12)		CDH74	8597	0	0	0	2	73	1062	2932	3312	1074	141	1	0	(12)
(13)		CDH80	1873	0	0	0	0	3	177	676	881	130	6	0	0	(13)
(14)	Wind	WSAvg	8.8	9.3	9.1	9.2	9.3	8.7	8.3	8.3	8.2	8.5	8.4	9.0	9.6	(14)
(15)	Precipitation	PrecAvg	24.5	1.7	1.3	1.6	1.9	1.9	1.7	0.9	2.3	3.1	3.9	2.4	2.2	(15)
(16)		PrecMax	41.4	8.2	6.0	4.9	6.9	9.1	4.2	3.7	7.0	10.9	16.2	11.5	10.0	(16)
(17)		PrecMin	15.9	0.0	0.0	0.0	0.1	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	6.4	1.7	1.4	1.2	1.2	1.6	1.1	1.1	1.6	2.6	3.3	2.6	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.3	66.6	69.5	72.7	79.2	87.7	89.2	92.2	85.8	80.6	71.8	66.0	(19)
(20)			MCWB	55.7	53.9	54.3	57.8	63.9	70.9	73.9	75.6	72.2	68.7	62.9	56.3	(20)
(21)		2%	DB	62.2	62.9	65.8	69.7	75.8	84.4	86.8	88.7	83.1	77.2	69.0	62.7	(21)
(22)			MCWB	53.1	52.8	54.6	58.3	63.5	71.7	74.6	74.8	72.5	68.8	61.2	55.4	(22)
(23)		5%	DB	59.4	60.3	63.0	67.5	73.5	81.9	85.2	86.1	81.2	75.3	66.8	60.9	(23)
(24)			MCWB	52.1	52.1	54.4	58.1	63.0	70.2	74.1	74.6	71.6	67.4	59.6	54.5	(24)
(25)		10%	DB	57.4	58.2	61.2	65.5	71.6	79.3	83.7	84.3	79.4	73.5	64.7	58.9	(25)
(26)			MCWB	51.3	51.4	53.9	57.3	62.8	69.3	73.5	74.3	70.2	66.1	58.3	53.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.5	56.9	59.1	62.4	67.9	76.3	78.9	80.0	76.9	72.8	65.4	59.3	(27)
(28)			MCD B	62.8	62.6	64.2	67.4	73.9	83.1	85.5	86.9	82.2	76.6	69.3	62.6	(28)
(29)		2%	WB	55.8	55.2	57.5	60.8	66.4	74.3	77.2	78.3	74.7	70.7	63.4	57.5	(29)
(30)			MCD B	60.1	59.9	62.3	66.2	72.4	81.1	84.2	85.3	80.8	75.3	67.7	61.3	(30)
(31)		5%	WB	54.2	53.8	56.4	59.6	65.1	72.4	76.0	76.9	73.3	69.1	61.5	55.7	(31)
(32)			MCD B	58.5	58.4	61.0	65.2	71.2	79.4	83.1	83.9	79.8	73.9	65.4	59.8	(32)
(33)		10%	WB	52.3	52.6	55.3	58.5	63.8	70.6	74.7	75.6	71.9	67.3	59.5	53.9	(33)
(34)			MCD B	56.7	57.2	59.8	64.0	69.9	77.7	82.0	82.6	78.3	72.4	63.7	58.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.2	14.3	13.9	13.1	12.6	12.3	12.0	12.6	13.1	13.0	14.1	14.0	(35)
(36)			MCDBR	15.5	16.2	16.0	15.7	14.5	14.1	13.2	13.9	13.8	14.1	15.0	14.8	(36)
(37)			MCWBR	10.8	10.9	10.2	9.6	9.0	9.1	9.3	9.9	9.2	9.7	10.9	10.7	(37)
(38)		5% WB	MCD BR	13.5	14.4	13.9	13.4	12.8	12.5	12.5	12.9	12.8	12.8	13.2	13.1	(38)
(39)			MCWBR	10.5	10.6	10.1	9.7	9.1	9.4	9.8	10.1	9.4	9.7	10.8	10.6	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.326	0.350	0.395	0.411	0.432	0.445	0.453	0.450	0.428	0.403	0.356	0.328	(41)	
(42)		taud	2.507	2.439	2.316	2.290	2.273	2.273	2.272	2.284	2.335	2.385	2.467	2.504	(42)	
(43)		Ebn at Noon	263	272	270	273	269	265	261	259	256	248	247	251	(43)	
		Edh at Noon	23	29	37	40	42	42	42	40	36	30	24	22		
(44)	All-Sky Solar Radiation	RadAvg	662	949	1374	1714	2029	2274	2248	1948	1481	1048	719	579	(44)	
(45)		RadStd	54	98	102	119	139	86	99	91	74	86	63	43	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+0.63	N/A	-2.63	N/A	N/A	N/A	N/A	N/A	+211	+160	
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149	+106	

Nomenclature: See separate page